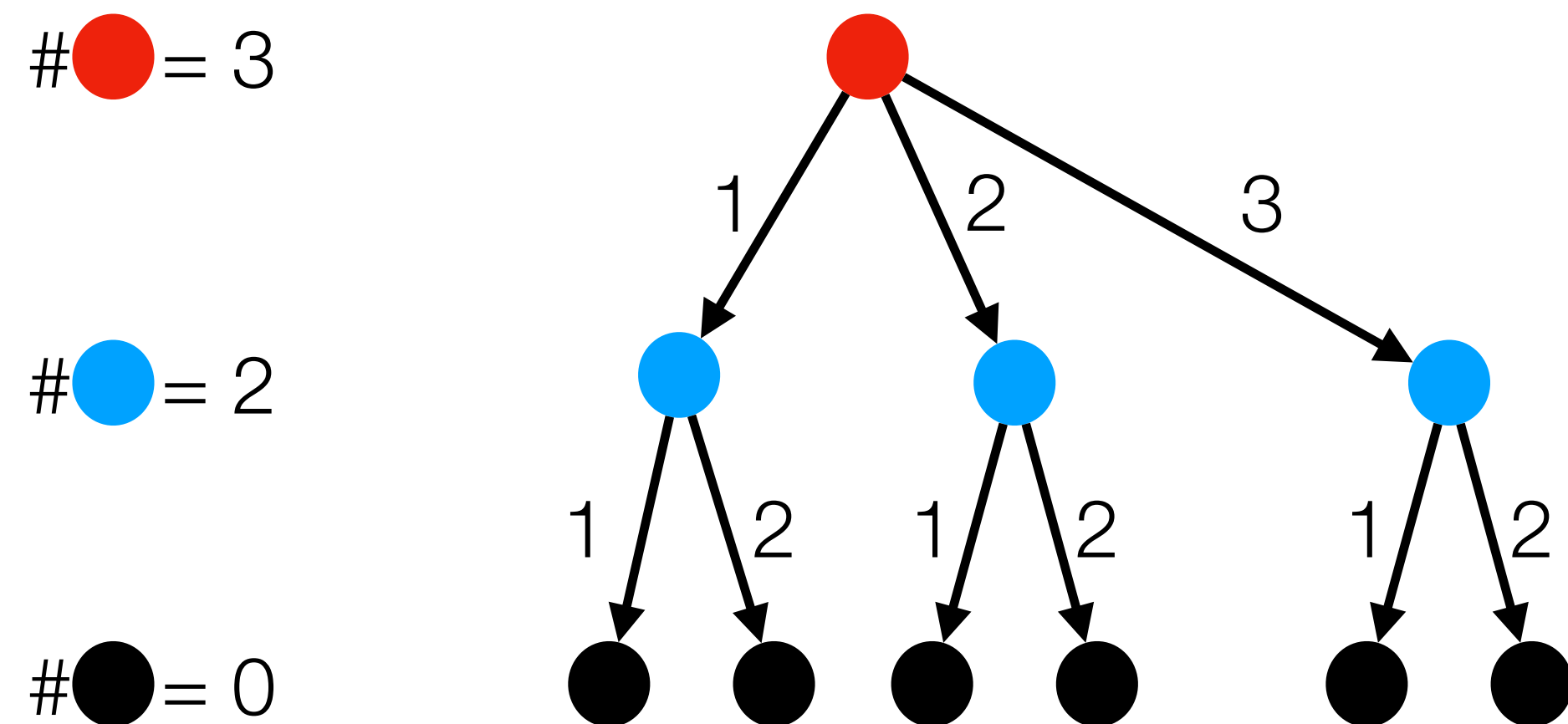


III. Trees

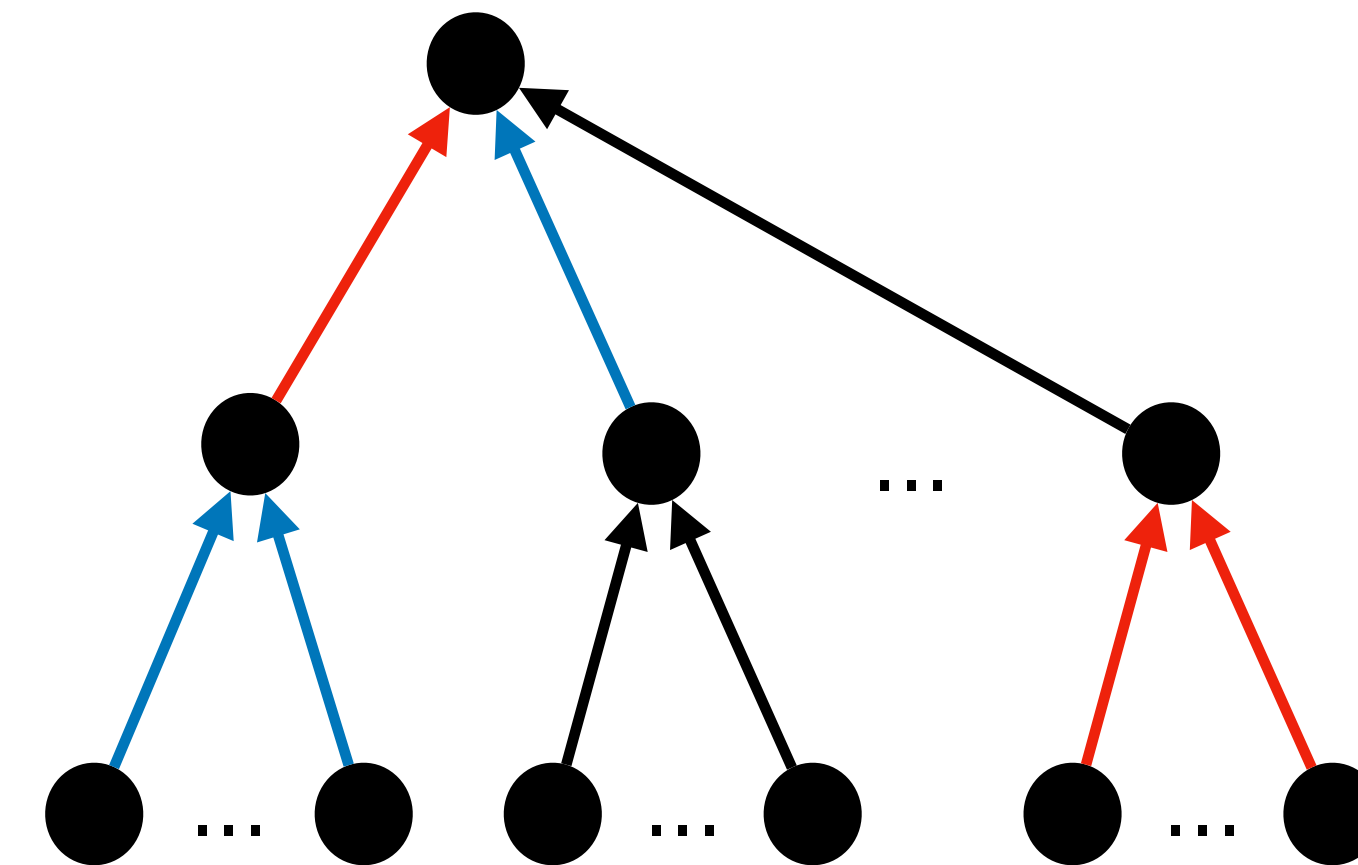
Radu Iosif (CNRS/VERIMAG)

Two Models



Ranked trees

number and order of children fixed by node labels



Unranked trees

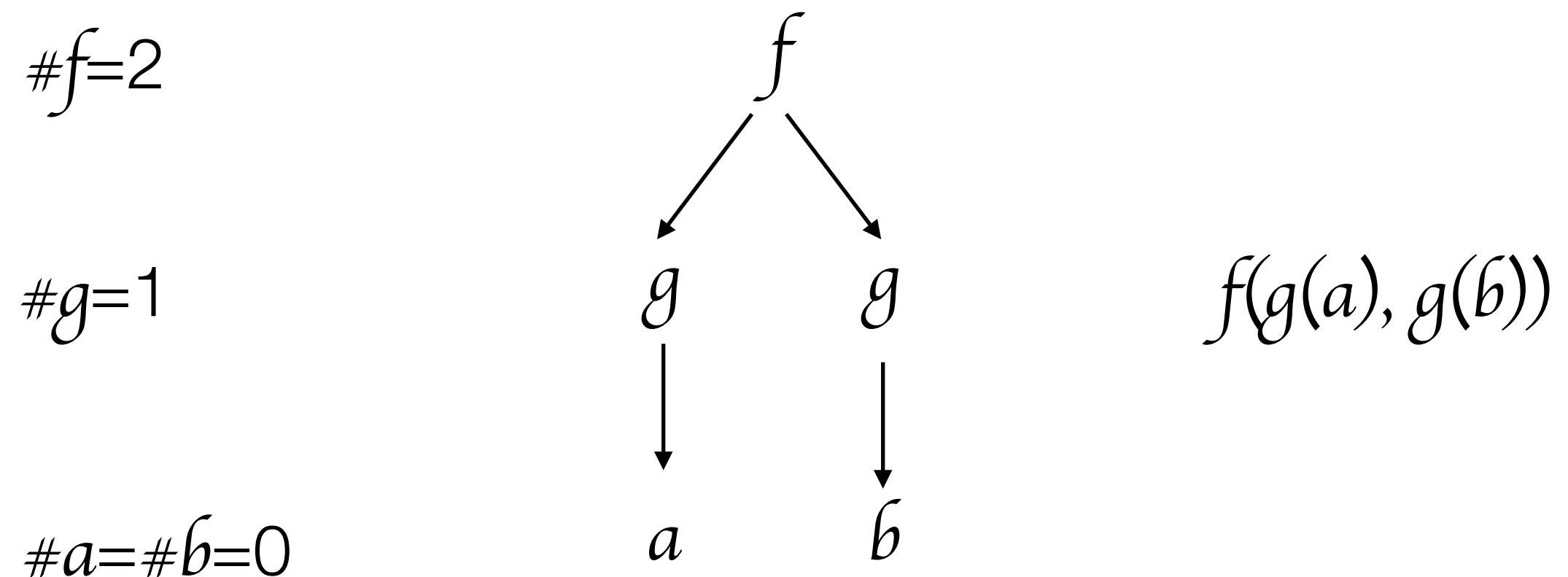
number and order of children not fixed

Recognisable Sets of Ranked Trees

$\Sigma \stackrel{\text{def}}{=} \text{finite alphabet of node labels}$

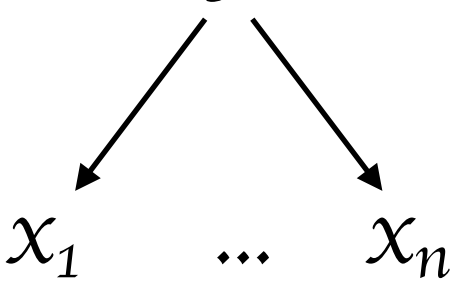
$\#a \stackrel{\text{def}}{=} \text{arity (rank) of } a \in \Sigma$

$\mathcal{T}(\Sigma) \stackrel{\text{def}}{=} \text{set of terms (ranked trees) over } \Sigma$



Recognisable Sets of Ranked Trees

$$\Sigma \stackrel{\text{def}}{=} \{f_1, f_2, \dots, f_n\}$$

$$f^{\mathcal{T}(\Sigma)}(x_1, \dots, x_n) \stackrel{\text{def}}{=} f$$


$$h(f^{\mathcal{T}(\Sigma)}(x_1, \dots, x_n)) = f^{\mathcal{A}}(h(x_1), \dots, h(x_n))$$

term algebra

finite algebra

$$(\mathcal{T}(\Sigma), f_1^{\mathcal{T}(\Sigma)}, f_2^{\mathcal{T}(\Sigma)}, \dots, f_n^{\mathcal{T}(\Sigma)}) \xrightarrow{h} (\mathcal{A}, f_1^{\mathcal{A}}, f_2^{\mathcal{A}}, \dots, f_n^{\mathcal{A}})$$

$\models$

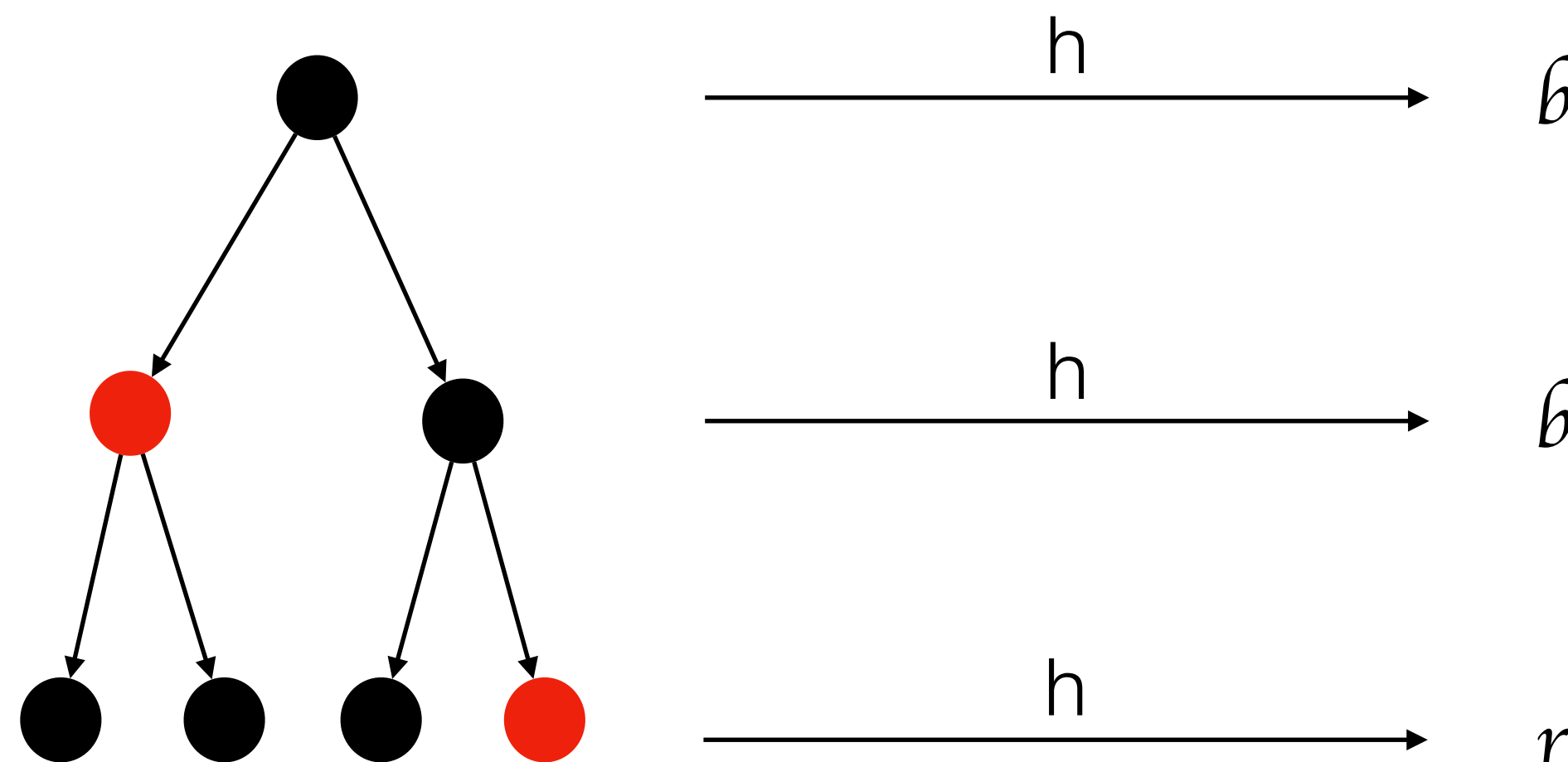
$\models$

$\mathcal{L}$

$=$

$h^{-1}(\mathcal{B})$

Recognisable Sets of Ranked Trees



$$f^{\mathcal{A}}(b, b) = r$$

$$f^{\mathcal{A}}(r, b) = f^{\mathcal{A}}(b, r) = f^{\mathcal{A}}(r, r) = b$$

Recognisable Sets of Ranked Trees

Theorem $\mathcal{L} \subseteq \mathcal{T}(\Sigma)$ is recognisable in the term algebra $(\mathcal{T}(\Sigma), \Sigma^{\mathcal{T}(\Sigma)})$

$\Leftrightarrow$

$\mathcal{L}$ is the language of a nondeterministic bottom-up tree automaton $A = (Q, F, \rightarrow)$

$$\begin{array}{ccc}
 \text{"}\Rightarrow\text{"} & (\mathcal{T}(\Sigma), f_1^{\mathcal{T}(\Sigma)}, f_2^{\mathcal{T}(\Sigma)}, \dots, f_n^{\mathcal{T}(\Sigma)}) & \xrightarrow{\quad h \quad} (\mathcal{A}, f_1^{\mathcal{A}}, f_2^{\mathcal{A}}, \dots, f_n^{\mathcal{A}}) \\
 & \textstyle\text{IU} & \textstyle\text{IU} \\
 & \mathcal{L} & = h^{-1}(\mathcal{B})
 \end{array}$$

$$A = (\mathcal{A}, \mathcal{B}, \rightarrow) \qquad (a_1, \dots, a_n) \xrightarrow{f} f^{\mathcal{A}}(a_1, \dots, a_n)$$

Recognisable Sets of Ranked Trees

Theorem $\mathcal{L} \subseteq \mathcal{T}(\Sigma)$ is recognisable in the term algebra $(\mathcal{T}(\Sigma), \Sigma^{\mathcal{T}(\Sigma)})$

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$$\begin{array}{ccc}
 \text{"}\Leftarrow\text{"} & (\mathcal{T}(\Sigma), f_1^{\mathcal{T}(\Sigma)}, f_2^{\mathcal{T}(\Sigma)}, \dots, f_n^{\mathcal{T}(\Sigma)}) & \xrightarrow{h(t) = \{q \mid t \rightarrow_A^* q\}} (2^Q, f_1^{\mathcal{A}}, \dots, f_n^{\mathcal{A}}) \\
 & \Downarrow & \Downarrow \\
 & \mathcal{L} & = h^{-1}(\{B \mid B \cap F \neq \emptyset\})
 \end{array}$$

$$f^{\mathcal{A}}(S_1, \dots, S_n) = \{q \mid \exists q_i \in S_i . (q_1, \dots, q_n) \xrightarrow{f} q\}$$

Logic for Ranked Trees

$\Sigma = \{f, g, a, \dots\}$ finite ranked alphabet

$child_i(x, y)$

y is the i -th child of x

$a(x)$

node x has label $a \in \Sigma$

Logic for Ranked Trees

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$\mathcal{L}(\varphi) = \{ t \in \mathcal{T}(\Sigma) \mid t \models \varphi \}$

language of a sentence

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language of a sentence

$\mathcal{L} \subseteq \mathcal{T}(\Sigma)$ is FO (MSO) definable iff $\mathcal{L} = \mathcal{L}(\varphi)$ for φ a FO (MSO) sentence

MSO-Definable Properties of Ranked Trees

$$closed(X) \stackrel{\text{def}}{=} \forall x . x \in X \wedge \mathbb{M}_{a \in \Sigma} (a(x) \Rightarrow \mathbb{M}_{1 \leq i \leq \#a} \underbrace{child_i(x)}_{\exists z' . child_i(z, z') \wedge z' \in X} \in X)$$

$$x \leq y \stackrel{\text{def}}{=} \forall X . (x \in X \wedge closed(X)) \Rightarrow y \in X$$

$$x \leq_{lex} y \stackrel{\text{def}}{=} x \leq y \vee \exists z . \mathbb{W}_{a \in \Sigma} a(z) \wedge \mathbb{W}_{1 \leq i < j \leq \#a} \underbrace{child_i(z)}_{\exists z' . child_i(z, z') \wedge z' \leq x} \leq x \wedge child_j(z) \leq y$$

$$root(x) \stackrel{\text{def}}{=} \forall y . x \leq y$$

$$leaf(x) \stackrel{\text{def}}{=} \forall y . x \leq y \Rightarrow x=y$$

MSO-definability = recognisability for ranked trees

Theorem [Doner] Let Σ be a finite ranked alphabet.

Each language $L \subseteq \mathcal{T}(\Sigma)$ is recognisable $\Leftrightarrow$ L is MSO-definable

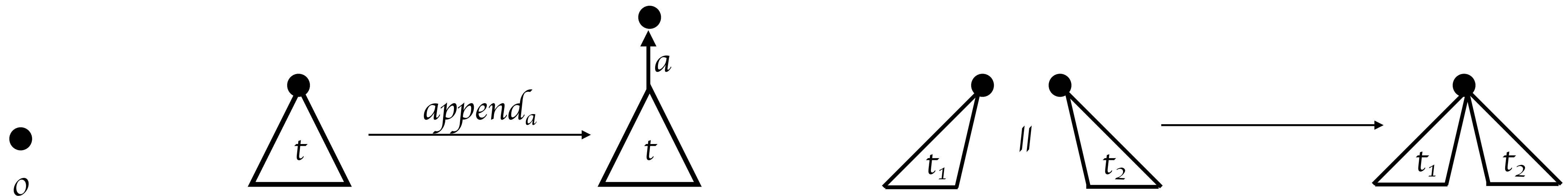
" $\Rightarrow$ " L is recognisable $\Rightarrow$ L is the language of a NBTA $\mathcal{A}$ with n states

$\exists X_1 \dots \exists X_n$. the sets $X_1 \dots X_n$ describe an accepting run of $\mathcal{A}$

Recognisable Sets of Unranked Trees

$\mathbb{A} \stackrel{\text{def}}{=} \{a, b, c, \dots\}$ = finite alphabet of binary edge labels

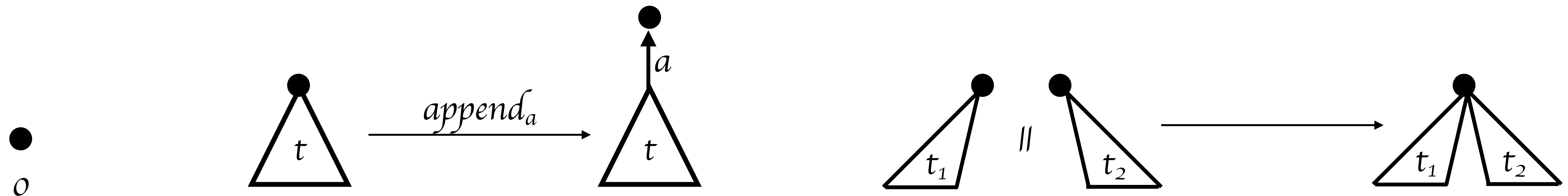
$\mathcal{U}(\mathbb{A}) \stackrel{\text{def}}{=} \text{set of unranked trees over } \mathbb{A}$



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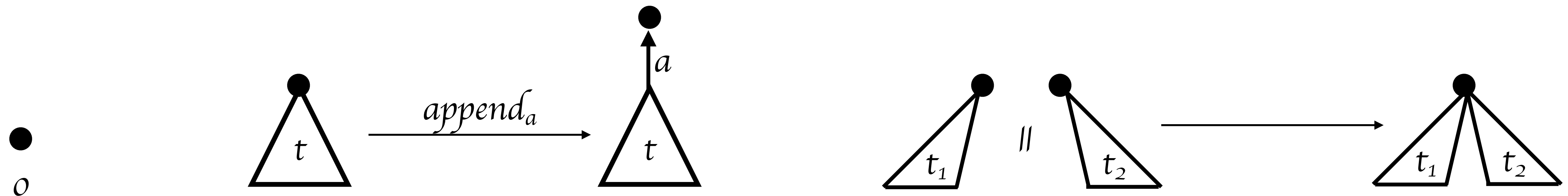


$(\mathcal{U}(\mathbb{A}), 0^{\mathcal{U}(\mathbb{A})}, ||^{\mathcal{U}(\mathbb{A})})$ commutative monoid

Recognisable Sets of Unranked Trees

$\mathbb{A} \stackrel{\text{def}}{=} \{a, b, c, \dots\}$ = finite alphabet of binary edge labels

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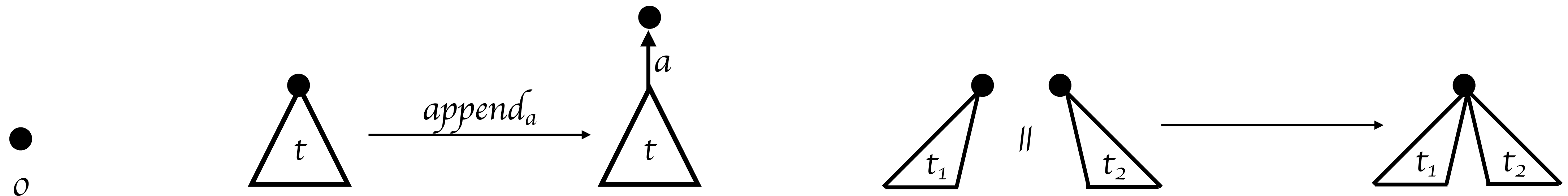


No suitable notion of automata exists for unranked trees

Recognisable Sets of Unranked Trees

$\mathbb{A} \stackrel{\text{def}}{=} \{a, b, c, \dots\}$ = finite alphabet of binary edge labels

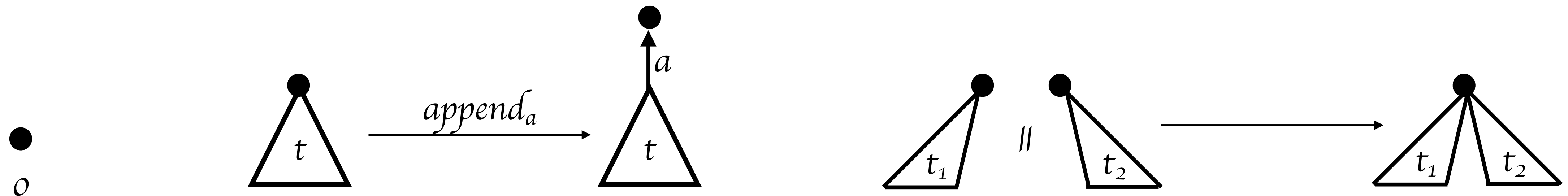
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unranked tree algebra

finite algebra

$$(\mathcal{U}(\mathbb{A}), 0^{\mathcal{U}(\mathbb{A})}, ||^{\mathcal{U}(\mathbb{A})}, \{append_a^{\mathcal{U}(\mathbb{A})}\}_{a \in \mathbb{A}}) \xrightarrow{h} (\mathcal{B}, 0^{\mathcal{B}}, ||^{\mathcal{B}}, \{append_a^{\mathcal{B}}\}_{a \in \mathbb{A}})$$

$\mathbb{U}$

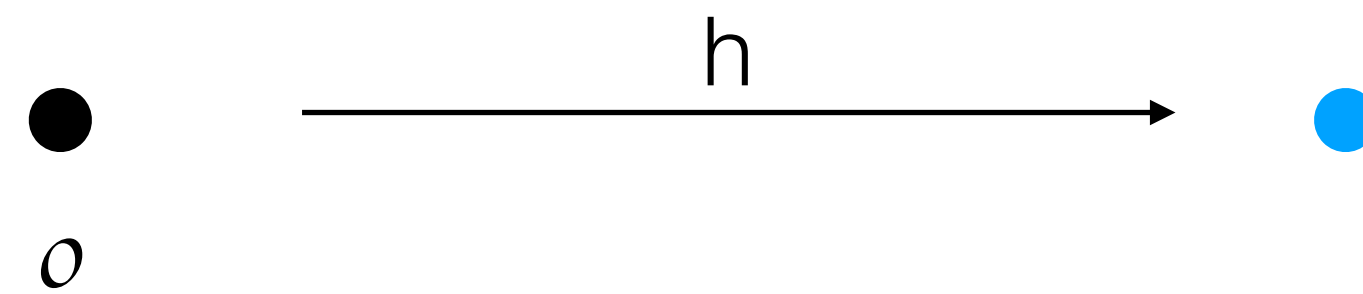
$\mathbb{U}$

$\mathcal{L}$

=

$h^{-1}(C)$

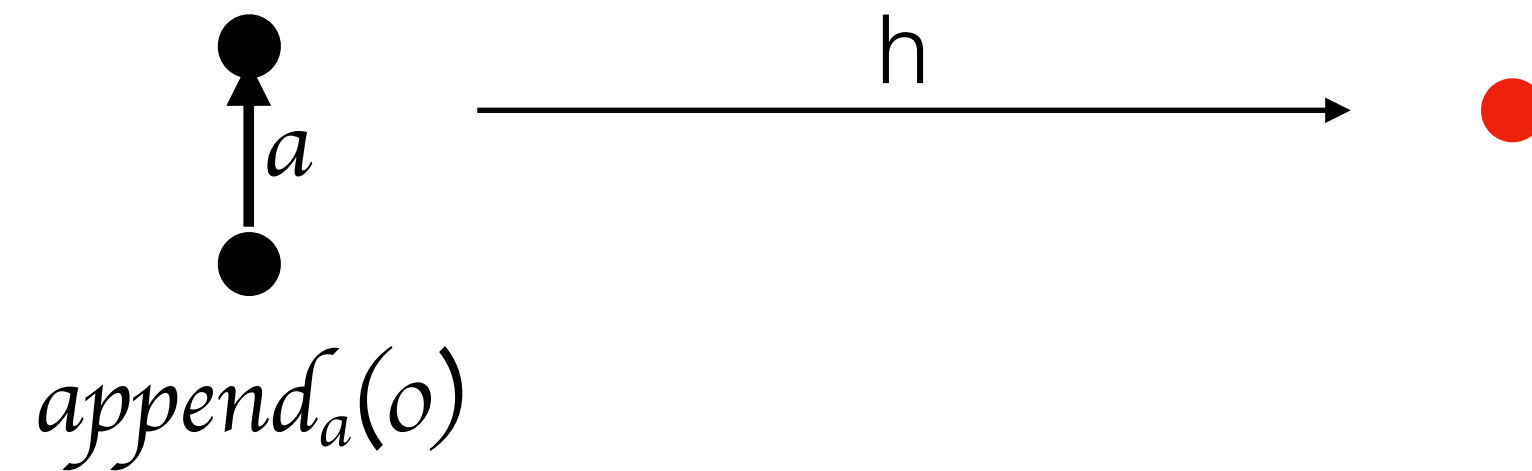
Recognisable Sets of Unranked Trees



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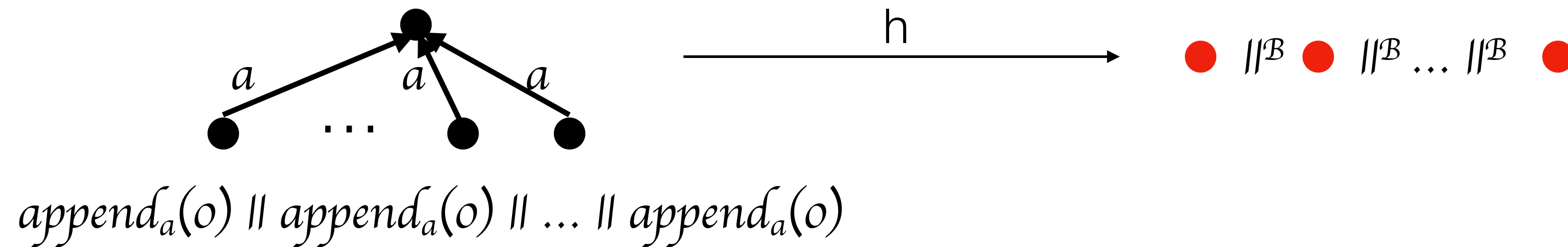
Recognisable Sets of Unranked Trees



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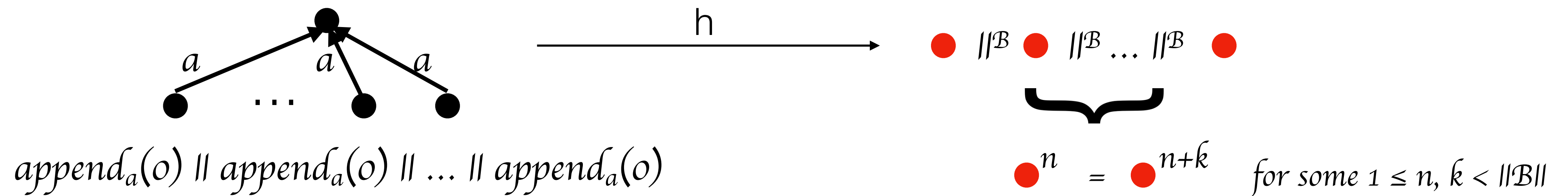
Recognisable Sets of Unranked Trees



$$(\mathcal{U}(\mathbb{A}), 0^{\mathcal{U}(\mathbb{A})}, \parallel^{\mathcal{U}(\mathbb{A})}, \{append_a^{\mathcal{U}(\mathbb{A})}\}_{a \in \mathbb{A}})$$

$$(\mathcal{B}, 0^{\mathcal{B}}, \parallel^{\mathcal{B}}, \{append_a^{\mathcal{B}}\}_{a \in \mathbb{A}})$$

Recognisable Sets of Unranked Trees



we need counting modulo $\parallel^{\mathcal{B}}$

$$(\mathcal{U}(\mathbb{A}), 0^{\mathcal{U}(\mathbb{A})}, \parallel^{\mathcal{U}(\mathbb{A})}, \{append_a^{\mathcal{U}(\mathbb{A})}\}_{a \in \mathbb{A}})$$

$$(\mathcal{B}, 0^{\mathcal{B}}, \parallel^{\mathcal{B}}, \{append_a^{\mathcal{B}}\}_{a \in \mathbb{A}})$$

Logic for Unranked Trees

$\mathbb{A} \stackrel{\text{def}}{=} \{a, b, c, \dots\}$ = finite alphabet of binary edge labels

$child_a(x, y)$

y is an a -child of x

$|X| = p \bmod q$

cardinality of X equals p modulo q

Counting Monadic Second Order Logic (CMSO)

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$\mathcal{L}(\varphi) = \{ t \in \mathcal{U}(\mathbb{A}) \mid t \models \varphi \}$

language of a sentence

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language of a sentence

$\mathcal{L} \subseteq \mathcal{U}(\mathbb{A})$ is (C)MSO definable iff $\mathcal{L} = \mathcal{L}(\varphi)$ for φ a (C)MSO sentence

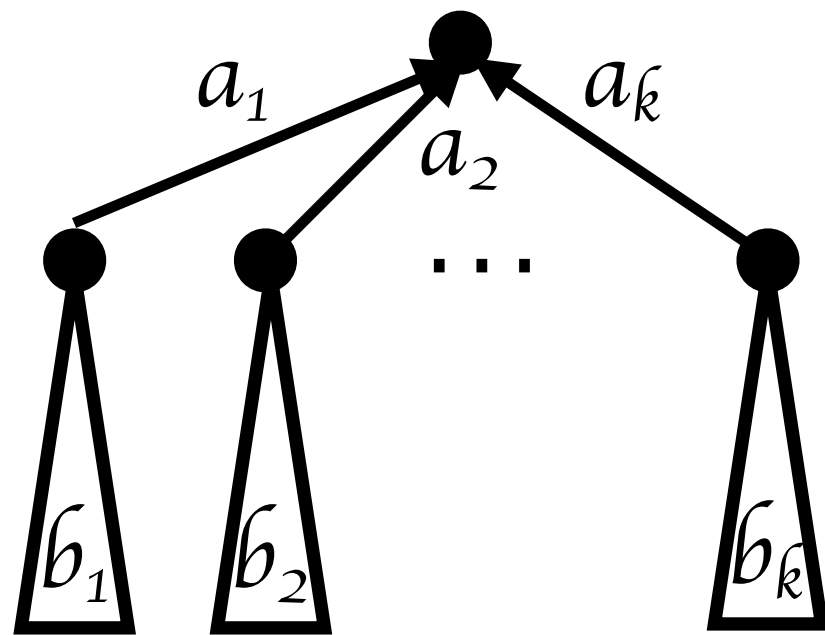
CMSO-definability = recognisability for unranked trees

Theorem [Courcelle] Let $\mathbb{A}$ be finite alphabet of edge labels.

Each language $\mathcal{L} \subseteq \mathcal{U}(\mathbb{A})$ is recognisable $\Leftrightarrow \mathcal{L}$ is CMSO-definable

$$\begin{array}{ccc}
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 \text{IU} & & \text{IU} \\
 \mathcal{L} & = & h^{-1}(C)
 \end{array}$$

$append_{a_1}(b_1) ||^{\mathcal{B}} append_{a_2}(b_2) ||^{\mathcal{B}} \dots ||^{\mathcal{B}} append_{a_k}(b_k)$
 $\left\{ \begin{array}{l} \text{are there exactly } 0 \leq n_{a,b} \leq ||\mathcal{B}|| \text{ terms } append_a(b) ? \\ \text{what is the value } n_{a,b} \text{ modulo } ||\mathcal{B}|| ? \end{array} \right.$

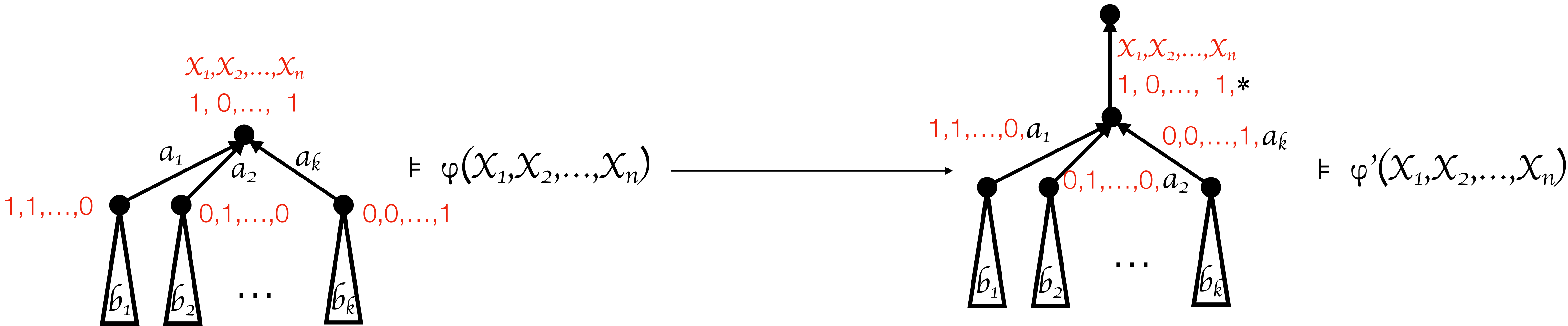


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Each language $\mathcal{L} \subseteq \mathcal{U}(\mathbb{A})$ is recognisable $\Leftrightarrow \mathcal{L}$ is CMSO-definable

" $\Leftarrow$ " $X = \emptyset$ $X \subseteq Y$ $child_a(X, Y)$ $single(X)$ $|X| = p \bmod q$
 $\forall y \in Y$ is a-child of $x \in X$ X is a singleton



CMSO-definability = recognisability for unranked trees

Theorem [Courcelle] Let $\mathbb{A}$ be finite alphabet of edge labels.

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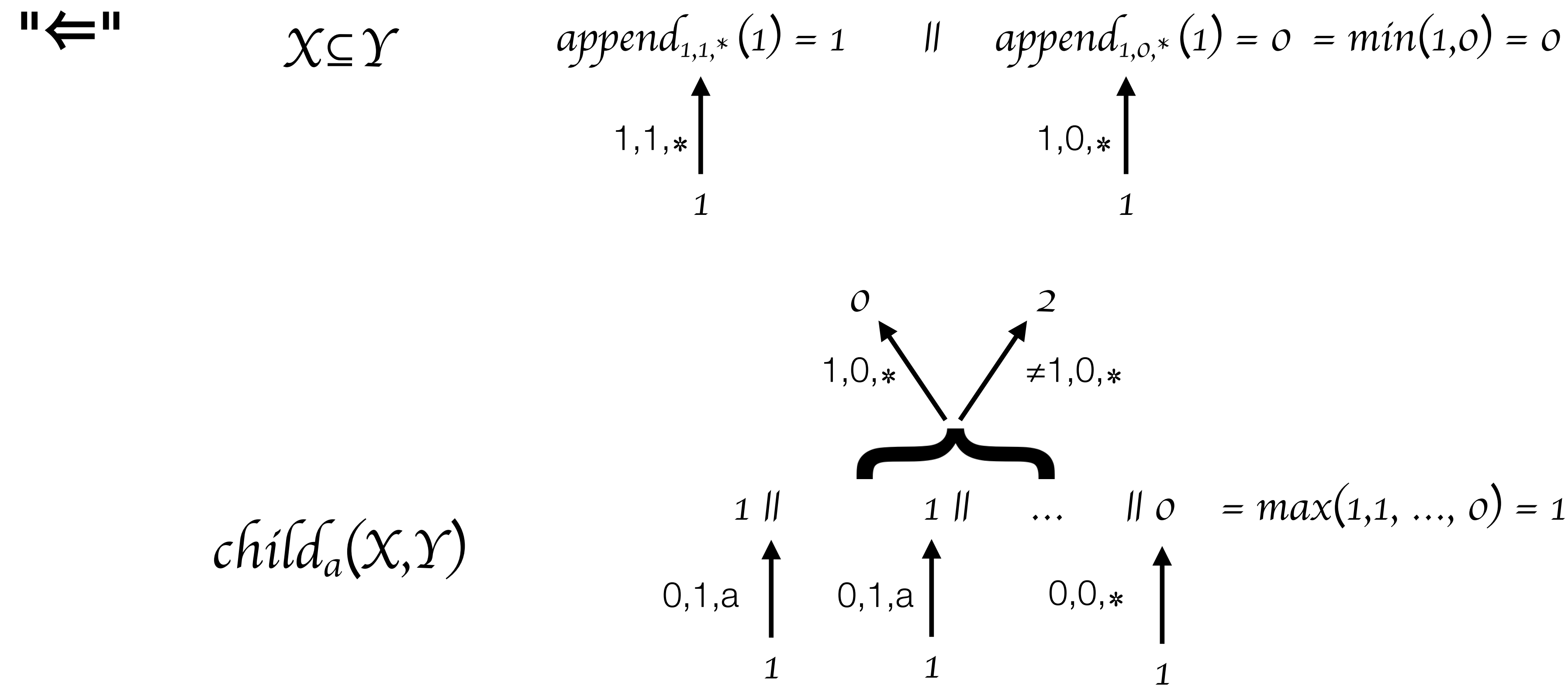
" $\Leftarrow$ "

$$\mathcal{X} \subseteq \mathcal{Y} \quad \begin{array}{c} \text{append}_{1,1,*}(1) = 1 \\ \uparrow \\ 1, 1, * \\ \uparrow \\ 1 \end{array} \quad \parallel \quad \begin{array}{c} \text{append}_{1,0,*}(1) = 0 = \min(1, 0) = 0 \\ \uparrow \\ 1, 0, * \\ \uparrow \\ 1 \end{array}$$

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" $\Leftarrow$ " $\varphi \wedge \psi, \neg\psi$ closure of recognizable sets under boolean operations

$\exists X. \varphi$ projection and powerset construction

Aperiodic Sets of Unranked Trees

$(M, 1, \cdot)$ monoid $m \cdot m = m$ idempotent

Idempotent Power Lemma Let $(M, 1, \cdot)$ be a **finite** monoid.

For each $m \in M$, the set $\{m, m^2, m^3, \dots\}$ contains exactly one idempotent $m^!$.

A finite monoid $(M, 1, \cdot)$ is **aperiodic** iff $m^! = m^! \cdot m$, for all $m \in M$

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A finite algebra $(\mathcal{A}, 0^{\mathcal{A}}, ||^{\mathcal{A}}, \text{append}_a^{\mathcal{A}}, \dots)$ is aperiodic iff $(\mathcal{A}, 0^{\mathcal{A}}, ||^{\mathcal{A}})$ is aperiodic

A language $\mathcal{L} \subseteq \mathcal{U}(\mathbb{A})$ is aperiodic iff it is recognised by a finite aperiodic algebra

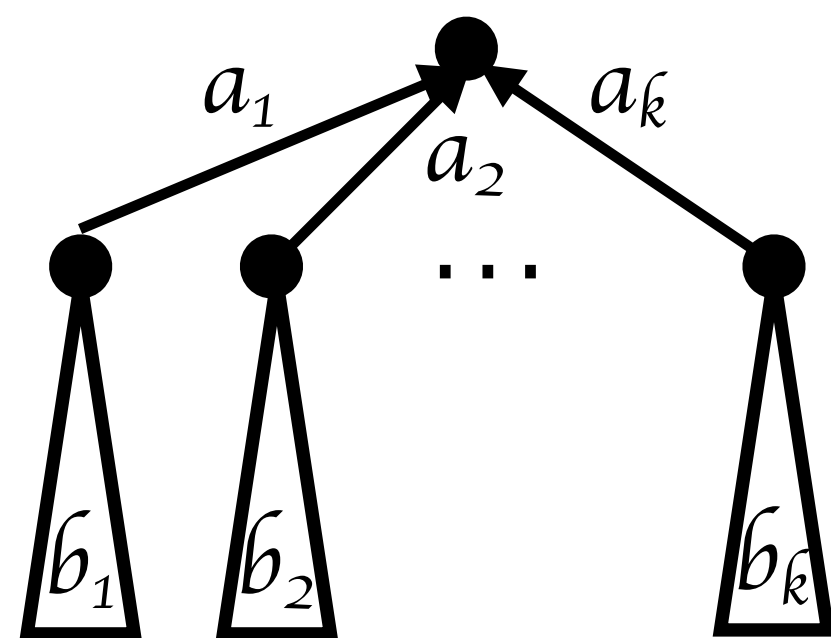
Aperiodic Sets of Unranked Trees

Theorem [Bojanczyk] Let $\mathbb{A}$ be finite alphabet of edge labels.

Each language $\mathcal{L} \subseteq \mathcal{U}(\mathbb{A})$ is aperiodic $\Leftrightarrow \mathcal{L}$ is MSO-definable

$$\begin{array}{ccc}
 \text{"}\Rightarrow\text{"} & (\mathcal{U}(\mathbb{A}), 0^{\mathcal{U}(\mathbb{A})}, ||^{\mathcal{U}(\mathbb{A})}, \{\text{append}_a^{\mathcal{U}(\mathbb{A})}\}_{a \in \mathbb{A}}) & \xrightarrow{h} (\mathcal{B}, 0^{\mathcal{B}}, ||^{\mathcal{B}}, \{\text{append}_a^{\mathcal{B}}\}_{a \in \mathbb{A}}) \\
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$\text{append}_{a_1}(b_1) ||^{\mathcal{B}} \text{append}_{a_2}(b_2) ||^{\mathcal{B}} \dots ||^{\mathcal{B}} \text{append}_{a_k}(b_k)$
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$\mathcal{L}$ is defined without modulo constraints

Aperiodic Sets of Unranked Trees

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" $\Leftarrow$ " same proof as before, except for the powerset construction in $\exists X. \varphi$

Lemma Let $(M, 1, \cdot)$ be a finite commutative aperiodic monoid.

Then $(2^M, \{1\}, \circ)$ is a finite aperiodic monoid, where $s \circ t \stackrel{\text{def}}{=} \{m \cdot n \mid m \in s, n \in t\}$

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Let $! \in \mathbb{N}$ be such that $m^! \cdot m = m$, for all $m \in M$

Let $k \stackrel{\text{def}}{=} (! + 1) \times ||M||$

For each $S \subseteq M$, $S^k = S^{k+1}$

Aperiodic Sets of Unranked Trees

Theorem [Bojanczyk] Let $\mathbb{A}$ be finite alphabet of edge labels.

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Corollary The language $\mathcal{L} \subseteq \mathcal{U}(\mathbb{A})$ of trees of height 1, having an even number of leaves is not MSO-definable

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Theorem [Courcelle] MSO and CMSO are equivalent on classes of structures having a total MSO-definable order.

Example 1. $(\Sigma^*, \leq)$ considering the successor model of the logic of words

Example 2. $(\mathcal{T}(\Sigma), \leq_{lex})$ considering the logic of ranked trees

Context-Free Sets

$\Sigma \stackrel{\text{def}}{=} \{f_1, f_2, \dots\}$ possibly infinite set of function symbols

$(\mathcal{A}, f_1^{\mathcal{A}}, f_2^{\mathcal{A}}, \dots)$ Σ -algebra

$\mathcal{L} \subseteq \mathcal{A}$ is **context-free** $\stackrel{\text{def}}{=} \mathcal{L} = \{t^{\mathcal{A}} \mid t \in \mathcal{T}\}$, $\mathcal{T} \subseteq \mathcal{T}(\mathcal{F})$ recognisable, $\mathcal{F} \subseteq \Sigma$ finite

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Theorem Each set of terms over a finite signature is recognisable $\Leftrightarrow$ it is context-free

Context-Free Grammars

Each tree automaton $A = (Q, F, \rightarrow)$ is equivalent to a context-free **grammar** $\Gamma = (Q, \leftarrow)$:

$$\underbrace{(q_1, \dots, q_n)}_A \xrightarrow{f} q$$

$$q \leftarrow f(\underbrace{q_1, \dots, q_n}_\Gamma) \leftarrow q, q \in F$$

$$\mathcal{L}^{\mathcal{A}}(\Gamma) \stackrel{\text{def}}{=} \{ t^{\mathcal{A}} \mid t \in \mathcal{L}(A) \}$$

Theorem [Filtering] Let $\Sigma \stackrel{\text{def}}{=} \{f_1, f_2, \dots\}$ be a set of function symbols and $(\mathcal{A}, f_1^{\mathcal{A}}, f_2^{\mathcal{A}}, \dots)$ be an algebra.

For each grammar Γ and set $\mathcal{L} \subseteq \mathcal{A}$ recognisable in $\mathcal{A}$, one can build a grammar Γ' such that $\mathcal{L}^{\mathcal{A}}(\Gamma') = \mathcal{L}^{\mathcal{A}}(\Gamma) \cap \mathcal{L}$